TASK SPECIFICATION: P1-05 PHASE P1 — FOUNDATIONS & DEVOPS INFRASTRUCTURE

Containerization, Docker Compose Orchestration & Health Checks

Git Branch: `feature/P1-05-docker-compose` · Target Pull Request: PR (feature/P1-05-docker-compose)

Core Objective & Technical Scope:
Deliver unified single-command full-stack containerization: Multi-stage Dockerfiles for backend and frontend (Nginx), PostgreSQL with persistent volumes, and health-check dependency ordering.

Priority Level	High	Complexity Rating	`M`
Execution Status	Done ✅	Target PR / Commit	`PR (feature/P1-05-docker-compose)`
Upstream Dependencies	P1-02, P1-03, P1-04		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Create multi-stage builder Dockerfile for Spring Boot with non-root runtime user.
2. Create multi-stage builder Dockerfile for React SPA with Nginx reverse proxy to /api.
3. Configure docker-compose.yml with postgres, backend, and frontend services on fbsu-network.
4. Configure container health checks with depends_on condition: service_healthy.
5. Author .env.example and comprehensive local development runbook.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/Dockerfile`
- `frontend/Dockerfile`
- `frontend/nginx.conf`
- `docker-compose.yml`
- `.env.example`
- `docs/runbooks/local-development.md`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- docker compose up boots all 3 containers to healthy state.
- Frontend serves UI on port 80; Backend responds on port 8080.
- Data persists in named volume across container restarts.

VERIFICATION & TEST COVERAGE STRATEGY

- docker build validation
- compose startup verification
- healthcheck validation
- volume persistence test